

Directly Controlling the Perceived Difficulty of a Shooting Game by the Addition of Fake Enemy Bullets

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ABSTRACT

Adjusting the balance between the player's game skill and the difficulty level is one of the most important factors to improve the player's engagement. However, it is still quite rare to find works that aim to straightly control the subjective difficulty perceived by the player. Our research question is whether or not it is possible to control the perceived difficulty just by adding enemy objects that do not raise the actual difficulty level. To investigate this issue, we designed a simple shooting game with two 'fake enemy bullets': Unreachable Bullets and Non-collisionable Bullets, which do not damage the player character. The experiment suggests that the non-collisionable bullet can efficiently increase the perceived difficulty level but the unreachable bullet does not. Such a study in novel techniques that can control the perceived difficulty without changing the actual difficulty could contribute to both research and practices in game design.

CCS CONCEPTS

• **Human-centered computing** → *Empirical studies in HCI; Empirical studies in interaction design.*

KEYWORDS

Difficulty adjustment, Perceived difficulty, Fixed actual difficulty, Game design

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1 INTRODUCTION

Balancing the game difficulty and player's skill is the topic getting increasing attention for enhancing the player's game experience. Based on the flow theory[2][3], the difficulty level should be designed carefully to continuously challenge the player but still match their skill. To reach the goal, Dynamic Difficulty Adjustment(DDA)

was proposed and became the most common method. Many researchers also have discussed and developed new techniques based on DDA system[10][7][5]. However, the current work all focus on tuning the parameter contributing to the real difficulty level to minimize the gap between the subjective difficulty and the real difficulty so that the user can enjoy the game.

Inspired by the Yanase's study[8], we found that he supposed to enhance the user's game experience with the fixed actual game difficulty. He claimed that the player would feel more skilled in the dodging game by adding some bullets that could never hit the player character. These bullets just appear on the screen visually but not contribute to the actual difficulty. Following Yanase's idea, we introduce the direct adjustment on the "perceived difficulty", a novel way of balancing the player's game skill and difficulty level. Here, we define the actual difficulty as the difficulty preset by the designer, which can be quantified as the given time limit, the number of enemies, etc. The perceived difficulty is the subjective assessment of the requirements of a task[4], which can only be measured by the player's subjective experience. It is still rare to find works aiming to investigate and directly adjust the subjective difficulty perceived by the player with the fixed actual difficulty.

Our research question is whether or not it is possible to control the subjective difficulty perceived by a player just by adding enemy objects that are visually present on the game screen but do not actually contribute to raising the difficulty level. To investigate this question, we introduce the "fake enemy bullets" that never damage the player character and design a simple shooting game. There are two kinds of fake enemy bullets: (1) Unreachable Bullets, which are shot to a direction that can never reach the player character, and (2) Non-collisionable Bullets, which do not make damage to the player character even when they collide with the player character. The unreachable bullet is designed according to Yanase's study[8]. These bullets look just like the regular enemy bullets in the shooting game. We have conducted a detailed user study with eighteen participants and assumed that the emergence of each fake enemy bullet can increase the perceived difficulty level. Based on the user study and hypothesis testing, we find that both kinds of fake enemy bullets can induce the fluctuation compared with the traditional shooting game that only contains the normal enemy bullets. However, only the addition of the non-collisionable bullet can obviously increase the perceived difficulty level. This result rejects the hypothesis about the unreachable bullet in the Yanase's study, but further proves the idea of adjusting the perceived difficulty with the fixed actual difficulty.

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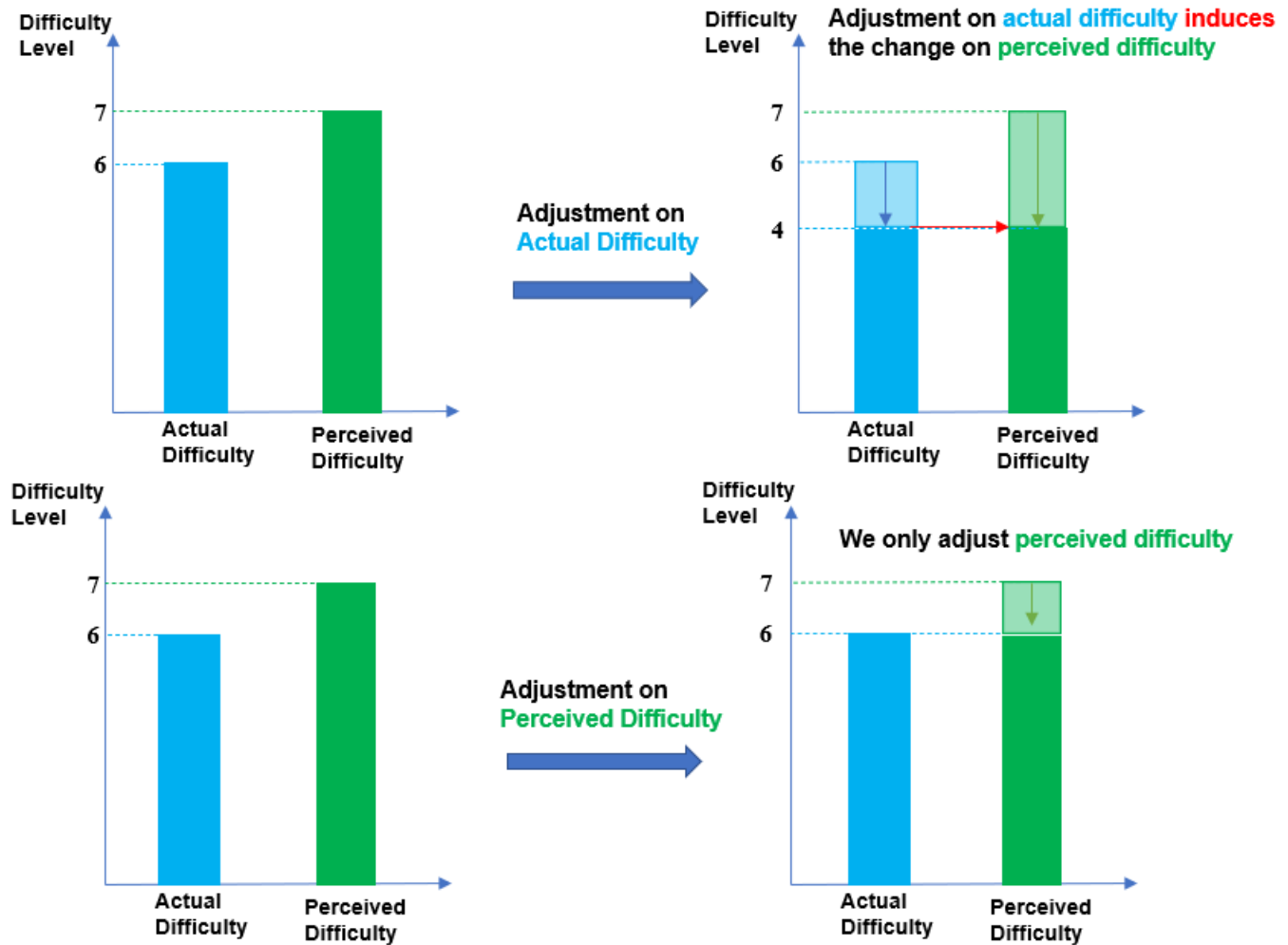


Figure 1: The differences between the adjustment on actual difficulty and perceived difficulty. The DDA system normally takes the data of the player's state and adjusts the actual difficulty to bridge the oversized gap between the actual difficulty and the perceived difficulty. Under this circumstance, both actual difficulty and perceived difficulty will change. However, our idea is only to adjust the perceived difficulty and keep the actual difficulty unchanged by adding the "fake enemy bullets" proposed in the paper.

While our work focuses on the shooting game, such a study in novel techniques that can control the perceived difficulty with the fixed actual game difficulty level could contribute to both research and practices in game design, as it provides alternative means to balance the player's game skill and the difficulty that players perceive, which could be helpful to enhance the player's engagement in gaming experience.

2 RELATED WORK

It is not the first time that the term "perceived difficulty" has been mentioned in the research concerned with the game difficulty adjustment. In 2009, Anthony et.al. discussed the challenge induced by the diversity of the player on adjusting the perceived difficulty in the video game[9]. As the development of machine learning, more and more DDA systems have been developed to help adjust

the perceived difficulty. The study of Daniel et.al. showed a machine learning model based on the procedural content to predict the player's perceived difficulty[6]. Su et.al. presented a framework of DDA system that is considered as an optimization problem[7]. Julian et.al. improved the DDA system by introducing the emotion-based adaptivity to better inform the system when to start DDA[5]. The advent of DDA system in different games indeed enhances the overall game experience efficiently[1]. However, the above work all tune the perceived difficulty by adjusting the parameter contributing to the actual difficulty, which are completely different from our proposed method.

There is only a few researches focusing on investigating the adjustment on the perceived difficulty with the fixed actual difficulty. Yanase proposed the similar idea in his work[8] in 2015. He has designed a dodging game. The player in the dodging game moves with

inertial force, and is expected to catch green reward. Every time the player catches one reward, one more new enemy is spawned and shoots one bullet to the player. There are three different types of enemy bullets:

- 1) The bullet is shot to the current position of the player.
- 2) The bullet is shot to the predicted position of the player, expecting the player keeps on moving to the same direction.
- 3) The bullet is shot to the position just near the player character that it can never hit the player, yet very close to the left/right limit where the player character may possibly reach.

The type 1 bullet is intended to give the motivation for the player to keep on moving from the current position. The type 2 bullet motivates the player to reduce the speed to avoid being hit by the type 2 bullet since the player is moving with inertial force. The addition of the type 3 bullet presents the idea of adjusting the perceived difficulty with fixed actual difficulty.

Yanase's research implicitly presented the idea of adjusting the perceived difficulty. However, he did not include any formal user study and just described the game idea only. Following Yanase's concept, we have designed a simple shooting game with the "fake enemy bullet" and conducted a serious user study to investigate our research question.

3 GAME DESIGN

3.1 Rule of The Game

To investigate the impact on the perceived difficulty induced by the addition of the fake enemy bullet, we design a shooting game, and Figure 2 shows our shooting game. The player character can only move at the bottom line of the screen with the left and right arrow keys. The enemy object will be continuously spawned from the top of the screen to the upper edge. Each enemy object will shoot one enemy bullet. The target in the game is randomly generated in the fixed area once a time. The next target is generated until the user shoots the previous target or the previous target has stayed for 3.5 seconds. The user can use the space bar to shoot the randomly generated target in the fixed area. To motivate the player to really aim at the target, the player can only shoot one bullet at a time. The next player bullet is ready until the previous player bullet collides with the target or goes out of the screen.

The player has unlimited life in the given time. The player character will be re-spawned at the bottom center of the screen after being hit with 1.5s invincible time. The red number on the top right corner records the number of time the player is hit by the enemy bullet. The green number in the top right corner records the number of time the player hits the target. The white number on the top left corner is the one-minute count-down timer.

3.2 Enemy Bullet

There are three different types of enemy bullets, the first one is the traditional enemy bullet, and the rest are the fake enemy bullet.

Table 1: Number of each bullet in each test

Test	Traditional Bullet	Unreachable Bullet	Non-collisionable Bullet
Test1	150	0	0
Test2	150	120	0
Test3	150	0	0
Test4	150	0	120

- **Traditional Enemy Bullet:** The traditional enemy bullet is shot to the current position of the player object and aims to hit the player object.
- **Fake Enemy Bullet, Unreachable Bullet:** The unreachable bullet is similar to the type 3 bullet in the Yanase's research. It goes to the position where the player character can never reach. We calculate the left and right impact point of the player character and the enemy bullet if the player keep moving to the left or right. Then we get a safe moving range of the player character corresponding to each enemy bullet. The unreachable bullet is shot to the border of the safe range so that it can never hit the player character.
- **Fake Enemy Bullet, Non-collisionable Bullet:** The non-collisionable is shot to the position of the player character. However, it will just go through the player character instead of making real damage.

4 EXPERIMENT

4.1 Experiment Setup

To investigate our research question, we hypothesize that the addition of each fake enemy bullet can increase the perceived difficulty level and conduct the experiment. We have arranged four one-minute tests with the mixture of the traditional enemy bullet and the fake enemy bullet. The first test contains only the traditional enemy bullet. The second test contains the mixture of the traditional enemy bullet and unreachable enemy bullet. The third test is the same as the first test. The fourth test contains the mixture of the traditional enemy bullet and non-collisionable enemy bullet. Each traditional enemy bullet is shot with a 0.4-second interval, and each fake enemy bullet is shot with a 0.5-second interval.

To avoid the bias induced by the participants' unfamiliarity with this game, they are first asked to go through four tests to get familiar with the game and then fill out the questionnaire in the second turn. We separate the questionnaire into page one and page two. The participants need to fill out page one of the questionnaire after finish playing the first two tests and reflect how difficult they think the second test is compared with the first test. After playing the third test and the fourth test, they are asked to fill out page two of the questionnaire to give feedback on the difficulty of the fourth test compared with the third test. In this way, we can analyze the independent consequences of each fake enemy bullets.

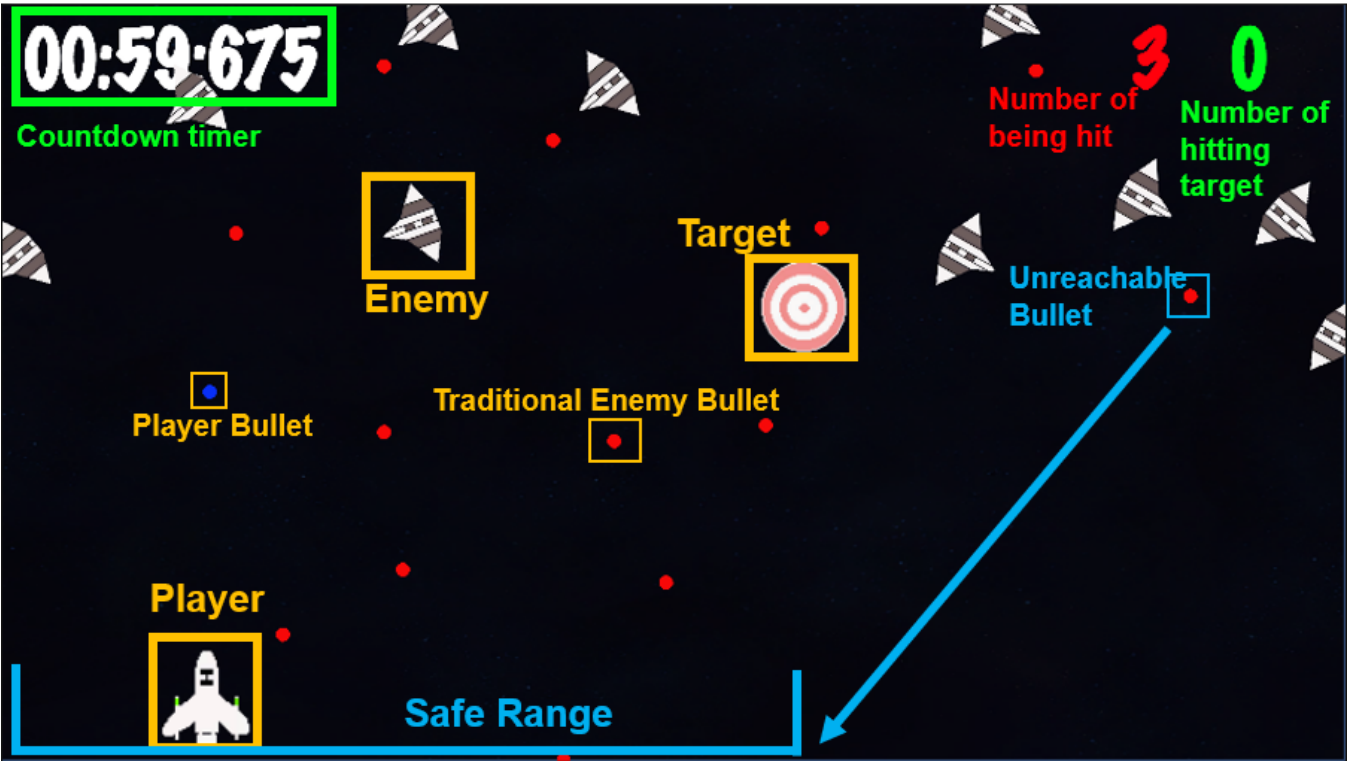


Figure 2: The game play screen. This is the second test containing the mixture of the traditional bullet and the unreachable bullet. The non-collisionable bullet is not shown in this figure. The non-collisionable bullet has the same appearance with the traditional bullet, but does not make real damage to the player character. The game objects in the above figure are different from the ones used in the experiment. The game objects are replaced to avoid the copyright issue here.

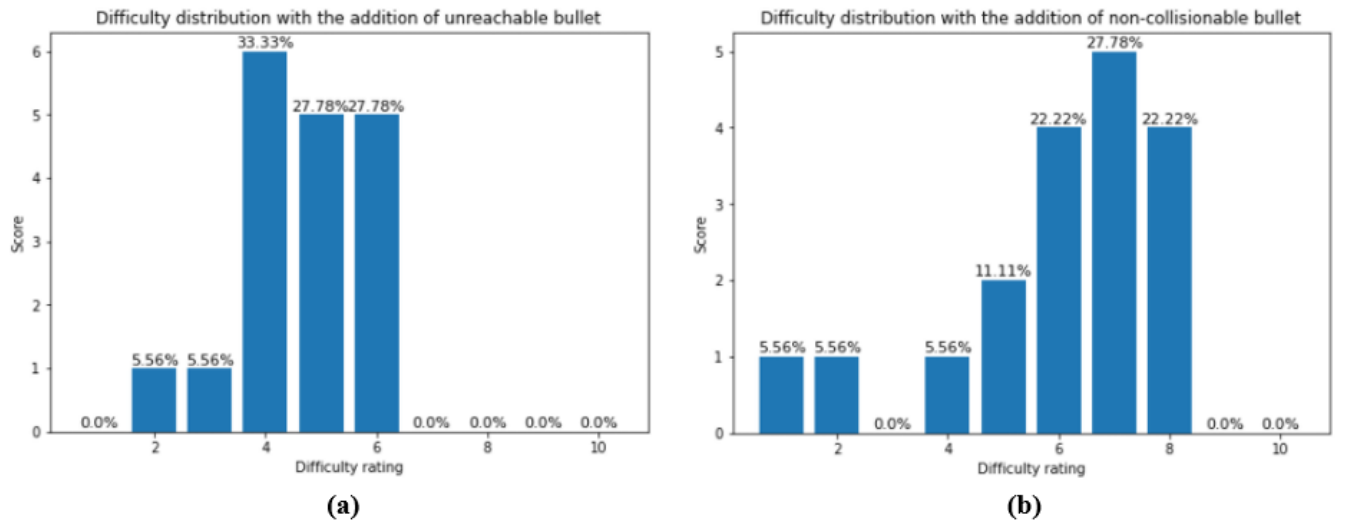


Figure 3: Evaluation on the perceived difficulty

Table 2: Hypothesis Testing

Type of Enemy Bullet	H_0	H_1	T-Value	P-Value	Conclusion
Unreachable Bullet	$\mu \leq 5$	$\mu > 5$	-1.243	0.1153	Fail to reject the $H_0(\alpha < 0.1153)$
Non-collisionable Bullet	$\mu \leq 5$	$\mu > 5$	2.121	0.0244	Reject the $H_0(\alpha > 0.0244)$

4.2 Result

Eighteen participants played our game and finished the user study. There are fifteen males and three females whose age ranges from 19 to 28. We set the difficulty level of the test1 and test3 to five as the standard, and we ask the participant to evaluate test2 and test4 compared with the previous test and evaluate the difficulty level. Figure 3 shows the distribution of the result. Both of the fake enemy bullets impact the perceived difficulty.

We adopt T-test($\alpha = 0.05$) to verify our hypothesis that the addition of each fake enemy bullet can increase the perceived difficulty level. Table 2 shows the detail of the hypothesis testing. Our Null Hypothesis is that the addition of the unreachable/non-collisionable bullet cannot increase the perceived difficulty level($\mu \leq 5$). The Alternative hypothesis is that the addition of unreachable/non-collisionable bullet can increase the perceived difficulty level($\mu > 5$). Based on the hypothesis testing, we find that the addition of the unreachable bullet actually cannot raise the perceived difficulty level, which violates the claim proposed in Yanase's paper[8]. However, the non-collisionable bullet can increase the perceived difficulty level based on the test data.

5 CONCLUSION

Our work investigates whether or not it is possible to control the subjective difficulty perceived by a player just by adding enemy objects that are visually present on the game screen but do not contribute to raising the actual difficulty level. We have designed the fake enemy bullet and a simple shooting game and conducted the user study. Based on the result, we find that the unreachable bullet cannot contribute to increasing the perceived difficulty level, which violates the claim in Yanase's paper[8]. However, the non-collisionable bullet is able to raise the perceived difficulty level, which gives a positive answer to our research question.

This research reveals a novel method of balancing the game difficulty and player's skill by directly adjusting the perceived difficulty level without changing the actual difficulty. Even if this research is based on a simple shooting game, it is still helpful for game development in industry and academia. Future work is involved with enlarging the sample size of the user study and pushing the current game design in the field of VR/AR game.

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